

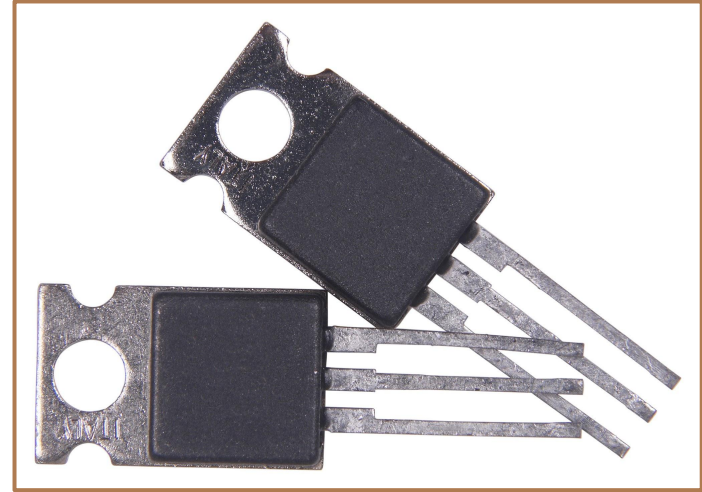
Field-Effect Transistor (FET)

Definition:

A **Field-Effect Transistor (FET)** is a three-terminal semiconductor device that controls the flow of electrical current using an electric field.

Key Characteristics:

- **Voltage-Controlled:** Operates by applying voltage to the gate, unlike current-controlled BJTs.
- **High Input Impedance:** Ensures minimal loading on previous circuit stages.
- **Three Terminals:** Comprised of the **Source**, **Drain**, and **Gate**.
- **Efficiency:** Known for low power consumption and high reliability.



Types of Field-Effect Transistors (FETs)

Junction FET (JFET)

The JFET is the simplest form of FET, utilizing a reverse-biased p-n junction to control the channel.

- Primarily used as voltage-controlled resistors.
- Excellent for low-noise amplifier circuits.
- Operates in depletion mode only.

MOSFET (Metal-Oxide-Semiconductor)

MOSFETs feature an insulated gate, allowing for even higher input impedance and better performance.

- Foundation of modern digital integrated circuits.
- Essential for microprocessors and memory chips.
- Supports both depletion and enhancement modes.

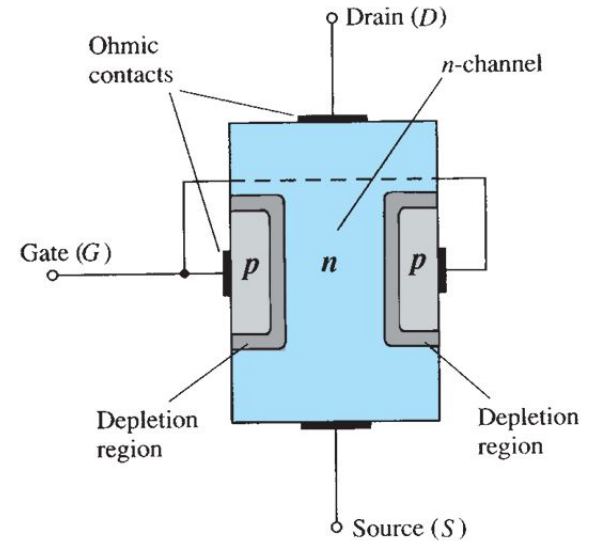
Junction Field Effect Transistor

Core Components

- **n-Channel:** The main conduction path made of n-type material.
- **p-type Gates:** Two embedded layers connected to form the **Gate (G)** terminal.
- **Drain (D) & Source (S):** Ohmic contacts at the top and bottom of the channel.

Key Concepts

- **Voltage Control:** The Gate terminal controls current flow between the Drain and Source.
- **Depletion Region:** Areas at the p-n junctions void of carriers; their size dictates the channel's width.
- **Analogy:** Functions like a **water spigot**, where the Gate acts as the valve controlling the "flow" through the channel.



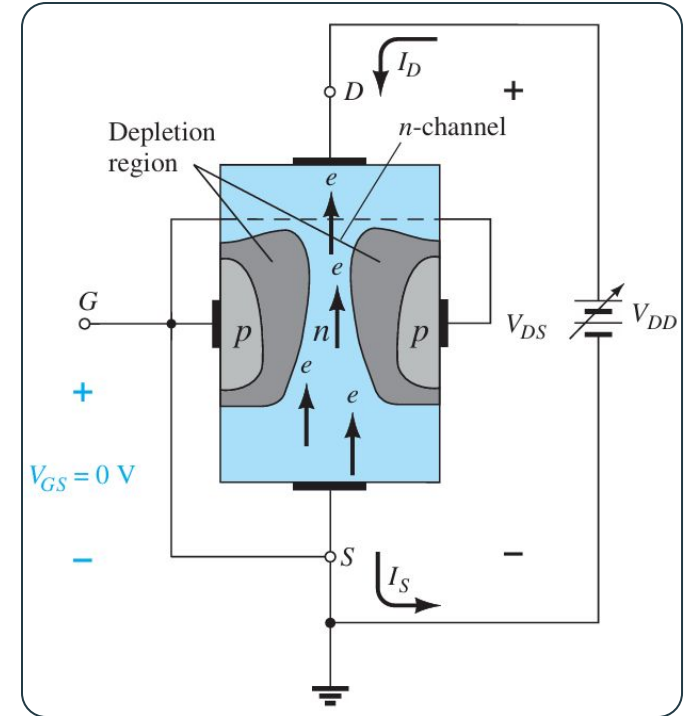
JFET Operation: $V_{GS} = 0V$, $V_{DS} > 0V$

Current Flow & Terminal States

- **$V_{GS} = 0V$:** Gate/Source are shorted at same potential.
- **Electron Flow:** V_{DS} pulls electrons from **Source to Drain**.
- **Equality:** Path is direct, so $I_D = I_S$.
- **Gate Current (I_G):** Always **0A**; junctions are reverse-biased.

Depletion Region Behavior

- **Non-Uniform Width:** **Wider at Drain**, narrower at Source.
- **Resistance Drop:** I_D creates a voltage gradient in the channel.
- **Gradient:** Top is $\sim 1.5V$; Bottom is $\sim 0.5V$ (relative to Gate).
- **Result:** Higher reverse bias near Drain narrows the path.



Internal Channel Dynamics during Bias

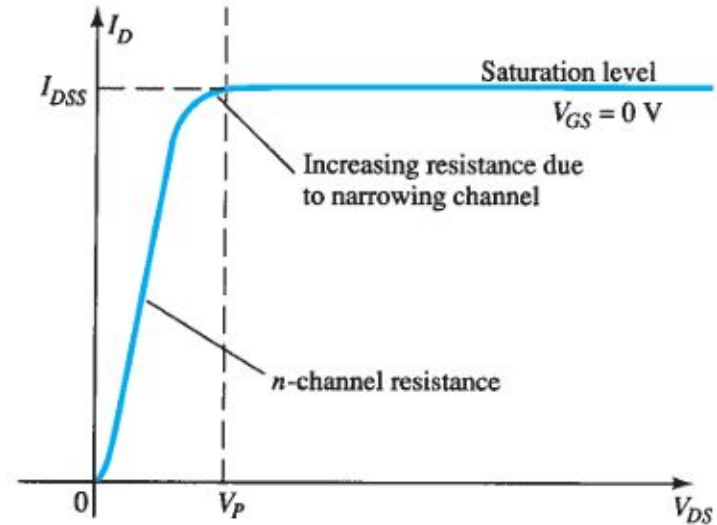
JFET Characteristics: Pinch-Off & Saturation

I-V Relationship ($V_{GS} = 0V$)

- **Ohmic Region:** At low V_{DS} , the channel acts as a fixed resistor; I_D increases linearly with V_{DS} according to Ohm's Law.
- **The Knee:** As V_{DS} rises, depletion regions widen, increasing channel resistance and causing the curve to flatten.
- **Pinch-Off (V_P):** The specific V_{DS} level where depletion regions nearly meet. Despite the name, the channel does **not** close completely.

Saturation Region (I_{DSS})

- **I_{DSS} (Drain-to-Source Saturation Current):** The maximum drain current for a JFET when $V_{GS} = 0V$.
- **Constant Current:** Beyond V_P , further increases in V_{DS} do not increase I_D . The device acts like a **constant current source**.



JFET Mechanics: Understanding Pinch-Off

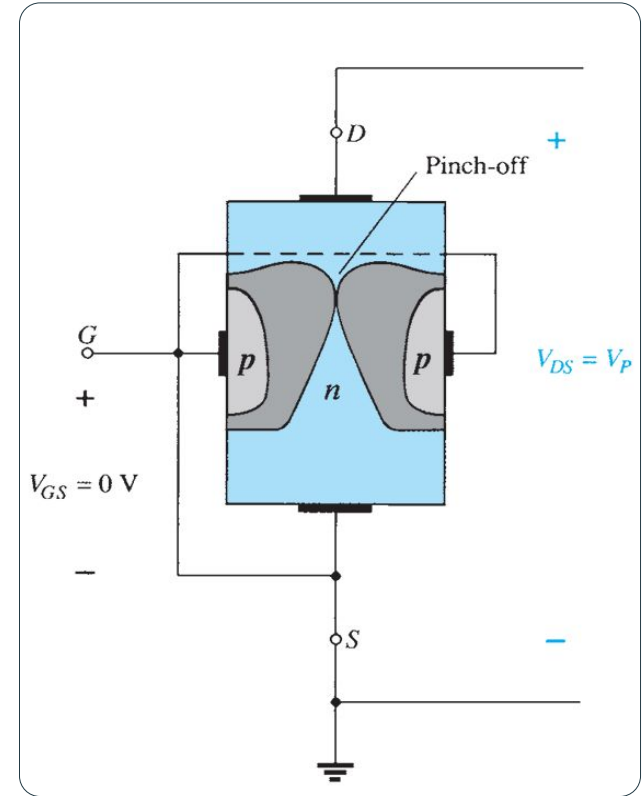
Pinch-off occurs when the **drain-to-source voltage (V_{DS})** causes depletion regions to widen and nearly meet near the drain terminal.

Core Principles

- **Pinch-off Voltage (V_P)**: The specific V_{DS} level where this state is reached while $V_{GS} = 0V$.
- **Saturation** : Marks the transition from the **Ohmic region** (linear current) to the **Saturation region** (constant current).

Key Characteristics

- **Channel Narrowing**: Reverse bias is strongest near the drain, creating a conduction "bottleneck" at the top.
- **Constant Current (I_{DSS})**: Current levels off at its maximum; further V_{DS} increases do not affect I_D .
- **Physical Reality**: The channel never fully closes to 0A; a thin path maintains a high-density electron stream.



Visual Representation of Channel Pinch-Off

Drain Current Equation

Shockley's Equation

The relationship is defined mathematically by:

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

I_{DSS} : Maximum drain current (at $V_{GS} = 0V$)

V_P : Pinch-off voltage (where $I_D = 0mA$).

V_{GS} versus I_D Using Shockley's Equation

V_{GS}	I_D
0	I_{DSS}
$0.3V_P$	$I_{DSS}/2$
$0.5V_P$	$I_{DSS}/4$
V_P	0 mA

Mathematical Analysis: Transfer Curve Sketch

Example 6.1: Sketch the transfer curve defined by $I_{DSS} = 12 \text{ mA}$ and $V_P = -6 \text{ V}$. (Ref: page 389)

Solution Strategy: Defining Plot Points

Four critical points are calculated using Shockley's Equation:

- **Point 1 (Saturation):** $I_D = 12 \text{ mA}$ at $V_{GS} = 0 \text{ V}$
- **Point 2 (Pinch-off):** $I_D = 0 \text{ mA}$ at $V_{GS} = -6 \text{ V}$
- **Point 3 (Mid-point):** At $V_{GS} = V_P/2 = -3 \text{ V}$,
 $I_D = I_{DSS}/4 = 3 \text{ mA}$
- **Point 4 (0.3 V_P):** At $I_D = I_{DSS}/2 = 6 \text{ mA}$,
 $V_{GS} \approx 0.3V_P = 0.3(-6\text{V}) = -1.8 \text{ V}$

These points define the parabolic transfer characteristics.

